

SOFTWARE ENGINEERING LAB-3

COMPONENT MODELLING & ARCHITECTURAL PATTERN SELECTION

Faculty Research Grant & Publication Tracker

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1. Objective

To evaluate different architectural styles, select the most appropriate architecture for the Faculty Research Grant & Publication Tracker, and design a UML Component Diagram showing the major components, interfaces, dependencies, and interactions between them.

2. Scenario Overview

The Faculty Research Grant & Publication Tracker is an automated portal for tracking sponsored research grants, faculty publication metrics, co-author approval workflows, and fund burn-up analytics. The main stakeholders are the Faculty Researcher and the Research Dean.

3. Key Requirements

- Track grant fund allocations, equipment procurement expenses, and remaining budgetary balance.
- Record faculty publications and track indexing status and citation metrics.
- Submit co-author details for approval and allow the Research Dean to review and approve them.
- View grant utilization and generate fund burn-up analytics.
- Maintain an immutable, tamper-evident audit ledger for financial approvals and status modifications.
- Preserve grant, publication, and approval data during normal operations.

4. Architectural Style Analysis

Architecture	Advantages	Disadvantages	Suitability
Layered Architecture	Clear separation of concerns; easy to understand; good for traditional applications.	Performance overhead; coupling between layers; difficult to scale individual components.	High
Microservices Architecture	Independent scaling; fault isolation; technology diversity; independent deployment.	Operational complexity; network latency; data consistency challenges.	Medium
Client-Server Architecture	Centralized control; simple deployment; easy data consistency.	Single point of failure; scalability bottleneck; limited fault tolerance.	Medium

5. Architecture Selection

Selected Architecture: Layered Architecture

We chose Layered Architecture for the Faculty Research Grant & Publication Tracker because the system has a clear separation between presentation, business logic, data management, and supporting services. This architecture provides a simple and maintainable structure for implementing grant management, publication tracking, co-author approval, utilization analytics, and audit processing.

Reason 1 — Separation of Responsibilities:

The platform contains distinct responsibilities — grant management, publication management, co-author approval, fund utilization, and analytics. Layered Architecture separates these into logical layers, making the system easier to understand, test, and maintain.

Reason 2 — Suitable for the Current System:

The tracker follows a centralized application structure with a user interface, business operations, a database, and supporting services. Layered Architecture fits this structure without introducing the operational complexity associated with independently deployed microservices.

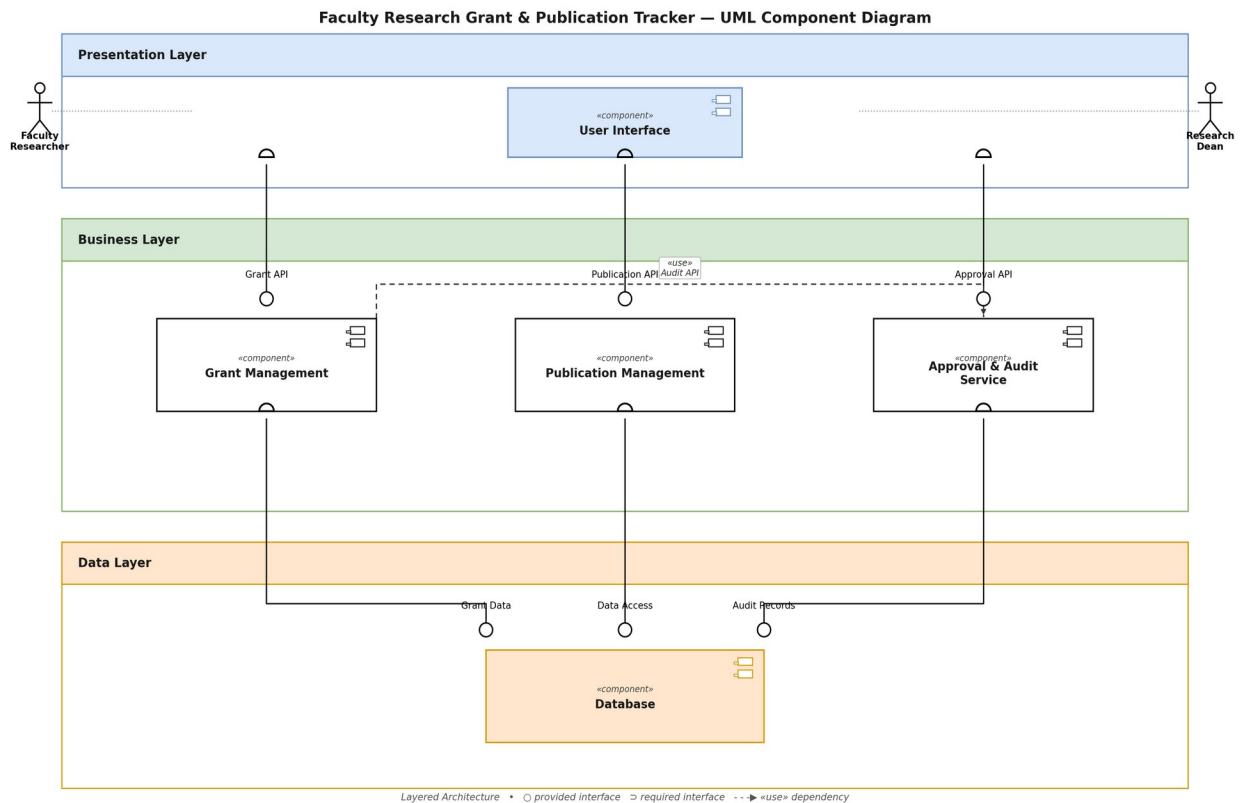
6. Security Advantage

Security-sensitive operations such as financial approvals and status changes are isolated in the business/service layer, while database access is controlled through defined interfaces. The Approval & Audit Service maintains a tamper-evident record of every financial and status modification, and is reached only through the Audit API rather than direct database access.

7. Performance Benefit

The business and data layers handle grant, publication, and analytics requests through efficient, dedicated data-access interfaces (Grant Data, Data Access, Audit Records) rather than a single shared channel. This reduces unnecessary interaction complexity and supports responsive operation under normal conditions.

8. UML Component Diagram



9. Component Interfaces and Interactions

Interface	Components	Purpose
Grant API	User Interface ↔ Grant Management	Grant allocation, expense, balance and utilization requests

Publication API	User Interface ↔ Publication Management	Publication, indexing and citation operations
Approval API	User Interface ↔ Approval & Audit Service	Co-author detail submission and Research Dean review/approval
Audit API («use»)	Grant Management → Approval & Audit Service	Records financial and status changes for auditing
Grant Data	Grant Management ↔ Database	Stores grant allocations, expenses and balances
Data Access	Publication Management ↔ Database	Stores and retrieves publication information
Audit Records	Approval & Audit Service ↔ Database	Stores audit and approval records

10. Data Flow

Faculty Researcher / Research Dean → User Interface → Grant Management / Publication Management / Approval & Audit Service → Database. Grant and publication information is persisted directly in the database, while co-author submissions route through the Approval & Audit Service, and every financial or status change made in Grant Management is additionally logged to the Approval & Audit Service over the Audit API before being written to the audit records store.

11. Conclusion

Layered Architecture is appropriate for the Faculty Research Grant & Publication Tracker because it provides clear separation between presentation, business logic, data management, and supporting services. It supports maintainability, controlled access to security-sensitive operations, reliable grant and publication data management, co-author approval workflows, and analytics, while the dedicated Audit API keeps every financial and status change traceable.